

BlueStock N100 — Analyst Guide

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1. Introduction

This guide explains how analysts can use the financial screening and dashboard system to evaluate companies, compare peers, analyse financial performance, and access company documents.

The system provides:

- A Streamlit-based analyst dashboard
- A company screener
- Company profile analysis
- Sector analysis
- Peer comparison
- Valuation and market-cap information
- Portfolio statistics
- Company documents
- PDF tearsheets
- REST API access for programmatic data retrieval

The underlying database contains financial and market information for **92 companies**.

2. System Overview

The project consists of the following major components:

Data Layer

The SQLite database contains the project's processed financial and market data, including:

- Companies
- Profit and loss
- Balance sheet

- Cash flow
- Financial ratios
- Sectors
- Peer groups
- Market capitalization
- Stock prices
- Documents
- Analysis
- Pros and cons

Analytics Layer

The analytics layer provides:

- Financial KPI calculations
- CAGR calculations
- Financial ratio analysis
- Company screening
- Peer analysis
- Portfolio statistics
- KMeans clustering

Dashboard

The Streamlit dashboard provides an interactive interface for analysts to explore the data.

REST API

FastAPI provides programmatic access to company, screener, sector, peer, valuation, portfolio and document data.

3. Starting the Application

3.1 Start the FastAPI Server

From the project root:

```
uvicorn src.api.main:app --port 8000
```

The API will be available at:

<http://localhost:8000>

FastAPI documentation is available at:

<http://localhost:8000/docs>

The /docs page provides interactive OpenAPI documentation and allows analysts or developers to test API endpoints.

4. Starting the Streamlit Dashboard

Start the dashboard using:

```
streamlit run dashboard.py
```

The dashboard normally runs on:

<http://localhost:8501>

The FastAPI server and Streamlit dashboard can be run simultaneously.

Recommended setup:

FastAPI → Port 8000

Streamlit → Port 8501

The dashboard obtains its data through the API.

5. Using the Company Screener

The screener allows analysts to filter and rank companies using financial criteria.

The screener supports filters such as:

- Minimum ROE
- Maximum Debt-to-Equity
- Minimum Free Cash Flow
- Sector
- Minimum 5-year Revenue CAGR
- Minimum 5-year PAT CAGR
- Maximum P/E

Example screening criteria:

Minimum ROE: 15%

Maximum Debt-to-Equity: 1.0

Minimum FCF: 100

Sector: IT

The screener returns companies satisfying the selected conditions.

The results contain the relevant company information and filter metrics so analysts can compare the shortlisted companies.

6. Screener Results

After applying the filters, analysts can review:

- Company ID
- Company name
- Sector
- ROE
- Debt-to-Equity
- Free Cash Flow
- Revenue CAGR
- PAT CAGR
- P/E
- Other available screening metrics

The results can also be downloaded as CSV for further analysis.

The CSV download should be checked to ensure that it is valid and well-formed.

7. Company Profile

The Company Profile screen provides detailed information about an individual company.

An analyst can select a company using its ticker.

The profile includes:

- Company information
- Sector

- Sub-sector
- Latest financial KPIs
- Financial performance
- Financial ratios
- Historical financial information
- Sector information
- Peer information
- Company documents

The profile is intended to provide a consolidated view of the company.

8. Profit & Loss Analysis

The P&L section provides historical profit and loss information.

Analysts can examine:

- Revenue
- Expenses
- Operating profit
- Profit after tax
- Other available P&L fields

Historical data can be filtered using the available year parameters.

The API supports:

from_year

to_year

for retrieving a selected period.

9. Balance Sheet Analysis

The Balance Sheet section provides historical balance-sheet information.

Analysts can examine items such as:

- Assets
- Liabilities

- Equity
- Debt
- Other available balance-sheet fields

Historical periods can be selected using the available year filters.

10. Cash Flow Analysis

The Cash Flow section provides historical cash-flow information.

Important areas include:

- Cash from operations
- Investing cash flow
- Financing cash flow
- Free cash flow
- Capital expenditure

Cash-flow information can be used alongside profitability metrics to evaluate the quality of reported earnings.

11. Financial Ratios

The Financial Ratios section provides calculated KPIs for the company across available years.

Important metrics include:

- Net Profit Margin
- Operating Profit Margin
- Return on Equity
- Debt-to-Equity
- Interest Coverage
- Asset Turnover
- Free Cash Flow
- Capex Intensity
- Dividend Payout Ratio
- Other calculated KPIs

Analysts can use the historical ratios to identify improving or deteriorating financial performance.

12. Sector Analysis

The Sector screen provides an overview of the available sectors.

The sector analysis includes:

- Number of companies
- Median ROE
- Median P/E
- Median Debt-to-Equity

Analysts can select a sector to view companies belonging to that sector.

This can be used to compare companies within the same industry and avoid comparing companies with fundamentally different business models.

13. Peer Analysis

The Peer section allows analysts to compare companies belonging to the same peer group.

Peer analysis provides:

- Companies in the peer group
- Percentile rankings
- Financial KPI comparisons
- Peer group averages
- Benchmark information

The peer comparison screen provides radar-style comparison data containing selected metrics for:

1. The selected company
2. Peer group average
3. Benchmark company

This helps analysts understand how a company performs relative to comparable companies.

14. Valuation and Market Capitalization

The Market Cap / Valuation section provides historical valuation information.

Available valuation multiples include:

- P/E
- P/B
- EV/EBITDA
- Dividend Yield

Historical valuation information is available for the specified period covered by the project.

Analysts can use this information to understand how market valuation has changed over time.

15. Portfolio Statistics

The Portfolio Statistics screen provides percentile statistics for the core KPIs across the company universe.

The statistics include:

- P10
- P25
- P50
- P75
- P90
- Mean
- Standard Deviation

These statistics help analysts understand where an individual company's KPI sits relative to the overall 92-company dataset.

For example, an analyst can determine whether a company's ROE is below, near, or above the median of the company universe.

16. Company Documents

The Documents section provides annual-report and other available company document links.

Each document record includes an `is_url_valid` indicator.

Analysts can use this section to access the source documents used for further company research.

17. PDF Tearsheets

The project generates company tearsheets in PDF format.

There should be one tearsheet for each company.

Tearsheets provide a consolidated company-level summary for analysis.

The generated tearsheets are stored under:

reports/tearsheets/

Analysts can open the relevant company PDF to review the company's financial information and analysis in a single document.

18. REST API Usage

The system provides REST API access through FastAPI.

Base URL:

`http://localhost:8000/api/v1`

Health Check

`curl http://localhost:8000/api/v1/health`

The health endpoint provides:

- API status
 - Database row counts
 - Uptime
 - Version information
-

19. Company API

Retrieve all companies:

`curl http://localhost:8000/api/v1/companies`

Retrieve a specific company:

`curl http://localhost:8000/api/v1/companies/TCS`

Retrieve P&L history:

```
curl http://localhost:8000/api/v1/companies/TCS/pl
```

Retrieve balance-sheet history:

```
curl http://localhost:8000/api/v1/companies/TCS/bs
```

Retrieve cash-flow history:

```
curl http://localhost:8000/api/v1/companies/TCS/cashflow
```

Retrieve financial ratios:

```
curl http://localhost:8000/api/v1/companies/TCS/ratios
```

Retrieve company documents:

```
curl http://localhost:8000/api/v1/companies/TCS/documents
```

20. Screener API

The screener can also be accessed programmatically.

Example:

```
curl "http://localhost:8000/api/v1/screener?min_roe=15"
```

Multiple filters can be combined.

Example:

```
curl "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1&sector=IT"
```

Invalid parameter values should return HTTP 400.

21. Sector API

Retrieve all sectors:

```
curl http://localhost:8000/api/v1/sectors
```

Retrieve companies in a sector:

```
curl http://localhost:8000/api/v1/sectors/IT/companies
```

22. Peer API

Retrieve companies in a peer group:

```
curl "http://localhost:8000/api/v1/peers/IT%20Services"
```

The endpoint returns the companies and their relevant percentile information.

23. Portfolio API

Portfolio statistics can be retrieved using:

```
curl http://localhost:8000/api/v1/portfolio/stats
```

The response contains the percentile statistics for the core KPIs.

24. API Documentation

Interactive API documentation is available at:

```
http://localhost:8000/docs
```

The OpenAPI specification is stored at:

```
docs/openapi.json
```

The OpenAPI specification can be used to understand available endpoints, parameters and response structures.

25. Troubleshooting

FastAPI Does Not Start

Run:

```
uvicorn src.api.main:app --port 8000
```

Check that:

- The virtual environment is active.
 - Required packages are installed.
 - The project is being run from the project root.
 - Port 8000 is not already occupied.
-

Streamlit Does Not Start

Run:

```
streamlit run dashboard.py
```

Check that:

- The virtual environment is active.

- Required dependencies are installed.
 - Port 8501 is available.
-

Dashboard Cannot Load Data

Check that FastAPI is running first:

`http://localhost:8000/api/v1/health`

Then verify the dashboard is running on:

`http://localhost:8501`

If the API is unavailable, dashboard data requests may fail.

API Returns 404

Check:

- The ticker is correct.
- The sector name is correct.
- The peer-group name is correct.
- The requested company exists in the database.

For example:

`/api/v1/companies/TCS`

should use a valid company ticker.

API Returns 400

A 400 response generally indicates invalid request parameters.

Check:

- Numeric filter values
 - Year values
 - Screener parameters
 - Sector names
 - Other query parameters
-

Database Errors

If an error such as:

no such table

appears, verify that the correct SQLite database has been created and populated.

Also verify that the application and tests are connecting to the same database file.

26. Recommended Analyst Workflow

A typical analysis can follow this sequence:

1. Open the Streamlit dashboard.
 2. Open the Screener.
 3. Apply financial filters.
 4. Review the shortlisted companies.
 5. Select an individual company.
 6. Review the Company Profile.
 7. Analyse P&L history.
 8. Analyse Balance Sheet history.
 9. Analyse Cash Flow history.
 10. Review Financial Ratios.
 11. Compare the company with its peers.
 12. Review sector-level statistics.
 13. Review historical valuation multiples.
 14. Check company documents.
 15. Generate or open the company's PDF tearsheet.
 16. Download screener results if required for further analysis.
-

27. Data and Analysis Scope

The dashboard and API are based on the processed project database containing **92 companies**.

The system is designed to support:

- Company screening
- Financial analysis
- Sector comparison
- Peer comparison
- Valuation analysis
- Portfolio-level statistics
- Document access
- Programmatic API access

The dashboard should be used as an analytical tool. Financial metrics should be interpreted together rather than relying on a single KPI.

28. Reference Files

Important project files include:

dashboard.py

src/

src/api/

docs/openapi.json

reports/tearsheets/

output/

reports/

The API documentation is available through:

<http://localhost:8000/docs>

The Streamlit dashboard is available through:

<http://localhost:8501>

29. Quick Reference

Component	Location
Streamlit Dashboard	dashboard.py
FastAPI Application	src/api/main.py
API Documentation	http://localhost:8000/docs
OpenAPI Specification	docs/openapi.json
Database	SQLite project database
Company Tearsheets	reports/tearsheets/
Cluster Labels	output/cluster_labels.csv
Portfolio Statistics	output/portfolio_stats.csv
Outlier Report	output/outlier_report.csv
Pytest Report	reports/pytest_report.html
Elbow Plot	reports/elbow_plot.png
Correlation Heatmap	reports/correlation_heatmap.png

30. Conclusion

The Analyst Guide provides the required instructions for using the financial screener, navigating the Streamlit dashboard, accessing company and financial information, comparing sectors and peers, retrieving data through the REST API, accessing documents and reviewing generated PDF tearsheets.

The system can be used through the dashboard for interactive analysis or through the REST API for programmatic access.